

HAIDA MAKOUANGOU

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EDUCATION

University of North Carolina at Charlotte

Aug 2023 - May 2026

Bachelor of Science, Computer Science - AI, Gaming and Robotics

Charlotte, NC

• **Achievements:** Cum Laude, Dean's List, Fall 2025

• **Coursework:** Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Software Engineering, Database Design & Implementation, Mobile Robotics, Web Application Development, Operating Systems & Networking, Computer Systems, Probability & Statistics, Linear Algebra

VIRTUAL JOB SIMULATIONS

Wells Fargo Software Engineering

July 16, 2026

- Understood relevant requirements for building a system to manage financial portfolios
- Drafted a visual representation of the data as an entity relationship diagram (ERD)
- Used the IntelliJ developer application to implement the ERD and published it to GitHub

Verizon Cloud Platform Job Simulation on Forge

November 2, 2025

- Completed a job simulation involving building a hypothetical new VPN product for Verizon's Cloud Computing team.
- Used command line Python to test whether Verizon's VPN met the cloud-native traits, i.e. redundancy, resiliency and least-privilege.
- Researched approaches to achieve application security and communicated insights in a PowerPoint Presentation.

GE Aerospace Explore Digital Technology virtual experience program on Forge

October 12, 2024

- Completed a Vue.js UI development simulation for the Frontend Development Team, demonstrating proficiency in Vue SFC Playground.
- Applied Vue.js skills by adding features like headings, images, and dynamic variable displays. Implemented a compute button for efficient user interactions.
- Drafted clear technical requirements for a feature, suggesting optimal order times for plane parts.
- Explored strategies to align technical specifications with business requirements, ensuring effective solutions.

PROFESSIONAL EXPERIENCE

Wells Fargo | *Engineering Associate*

Jul 2022 - Apr 2024

- Managed and updated 100+ ServiceNow tickets monthly to support infrastructure changes, CI/CD automation, and issue triage within IT governance and change management processes.
- Collaborated with Data Center Hosting Services to coordinate hardware decommissioning, validate asset inventory records, and document implementation requirements for enterprise governance and asset lifecycle controls.
- Supported software testing and vulnerability remediation through production troubleshooting, documenting infrastructure changes, test results, and performance metrics in Confluence and Microsoft Excel.

Software Engineer Intern- Team Load Balancer

Jan 2022 - Jul 2022

- Supported the Load Balancer Engineering team by managing ServiceNow ITSM change tickets, validating implementation details for network and configuration updates, and maintaining accurate technical documentation.
- Performed network connectivity troubleshooting and validation using command-line tools, including ping, nslookup, tracer/trace route, IP address verification, DNS, and CNAME record validation, to support reliable enterprise infrastructure changes.
- Collaborated with Load Balancer Engineering teams to support application availability, traffic distribution, and reliable enterprise network operations.

PROJECTS

AI Job Market ML Project | Python, scikit-learn, Panda

Fall 2025

UNC Charlotte

- Built a supervised machine learning model to predict whether demand for a job will increase or decrease between 2024 and 2030
- Cleaned and analyzed more than 30,000 records, then trained Logistic Regression and Random Forest models.
- Evaluated predictions using F1 score and a confusion matrix.

Trade Reconciliation System | Java, Spring Boot, Spring Data JPA, H2, Maven, AWS

September 2026

<http://trade-reconciliation-haida.us-east-1.elasticbeanstalk.com/>

- Built a financial application that compares internal and external transactions to identify matches, amount differences, and missing records.
- Developed REST API endpoints and used Spring Data JPA with an H2 database to store and reconcile trade records.
- Added CSV import and JUnit tests; deployed the application to AWS Elastic Beanstalk.

AI Study Assistant | Python, Flask, React, FAISS, Embeddings

Summer 2026

<https://ai-study-assistant-lac.vercel.app/>

- Built a retrieval-augmented learning assistant that lets users upload study materials and ask questions grounded in their documents.
- Used embeddings and FAISS to retrieve relevant content, with a Flask backend and React frontend.
- Added tools to generate summaries, quizzes, and flashcards from study materials.

TRAINING, CERTIFICATIONS & COMMUNITY

- Year Up, Application Development · Per Scholas, Software Engineering · CodePath, AI Engineering
- CompTIA Security+ · Microsoft Azure Fundamentals (AZ-900) · Google IT Support Professional Certificate
- Member: ColorStack and Women in Tech · Participated in an event on implementing generative AI in the financial sector

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, TypeScript, HTML/CSS, C++, SQL (Postgres)
- **Frameworks:** React, Node.js, Flask, Junit, Material-UI, FastAPI, Express.js, Spring Boot, Bootstrap, Spring Data JPA
- **AI & Data:** Pandas, NumPy, Matplotlib, Scikit-learn, Hugging Face, FAISS, retrieval-augmented generation (RAG)
- **Databases & Cloud:** PostgreSQL, MySQL, MongoDB, H2, AWS Elastic Beanstalk, Azure, Vercel, Render
- **Tools & Systems:** Git, VSCode, PyCharm, IntelliJ, Agile, ServiceNow, Confluence, Postman, Linux, Unity 2D, DOTween